FIITJEE **JEE (Advanced)-2018**

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - IV

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 186**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into Two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01– 05, 19 – 23, 37 - 41)*** *contains 15 multiple choice questions which have* ***one correct*** *answer. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (06 – 13, 24 – 31, 42 - 49) *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer.*

   Partial Marks **+1** for each correct option provided no incorrect options is selected.
2. ***Section-C (14 – 18, 32 – 36, 50 – 54)*** *contains 15 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **5 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

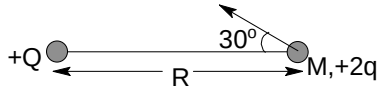
1. In a screw gauge there are 100 divisions in the circular scale and each main scale division is of 1 mm. When there is no gap between the jaws, 97th division coincides with the main scale zero and zero of main scale is not visible. While measuring the diameter of a ball, the circular scale is between the 3mm mark and the 4mm mark such that the 76th division of the circular scale coincides with the reference line. Then,
   (A) the least count of the micrometer is 0.01 cm (B) the zero error is -0.04 mm
   (C) the main scale reading is 4 mm (D) the diameter of the ball is 3.79 mm

2. A ball is projected from a point 10 m away from a wall. If the initial velocity with which the ball can be projected is 15 m/s, then in order to hit the wall at the highest possible point the angle with the horizontal that it must be projected is
   (A) 45 (B) 90
   (C) $\tan^{-1} 2.25$ (D) $\tan^{-1} 1.5$

3. A light thread is wound four times around a pulley disc of diameter 30 cm and mass 5 kg. We want to unwind the thread by pulling it in 1 s. The minimum constant force required to do so is:
   (A) 12.25 N (B) 18.85 N
   (C) 24.5 N (D) 37.7 N

Space for Rough work

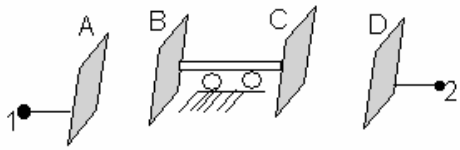
4. In the diagram shown, the charge +Q is fixed. Another charge +2q, is projected from a distance R from the fixed charge. The minimum separation between the two charges when the velocity becomes $\frac{1}{\sqrt{3}}$ times of the projected velocity, is (Assume gravity to be absent)



(A) $\frac{\sqrt{3}}{2}R$ (B) $\sqrt{3}R$

(C) $\frac{1}{2}R$ (D) $4R$

5. A, B, C and D are large conducting plates kept parallel to each other. A and D are fixed. Plates B and C, connected to each other by a rigid conducting rod, can slide over frictionless rails as shown. Initially the distance between plates A and B is the same as that between plates C and D. If now the rod (along with plates B and C)is slightly moved towards right, the capacitance between the terminals 1 and 2.



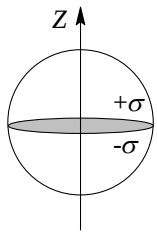
(A) remain unchanged (B) increases
(C) decreases (D) nothing can be said

Space for Rough work

(One or More than one correct type)

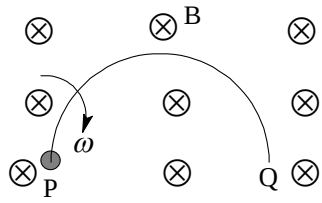
This section contains **EIGHT** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

6. A hollow, insulating spherical shell has a surface charge distribution placed upon it, such that the upper hemisphere has a uniform surface charge density $+\sigma$, while the lower hemisphere has a uniform charge density $-\sigma$,as shown in the figure. Their interface lies in x-y plane. Which of the following statement(s) is/are correct?



(A) The field at all points of x-y plane within the sphere points in the $-ve$ z-direction
(B) All points of the x-y plane within the sphere are equipotential
(C) The field at all points on z-axis outside the sphere point along positive z-direction
(D) The field at points on z axis which are on either side of origin outside the sphere is in opposite directions.

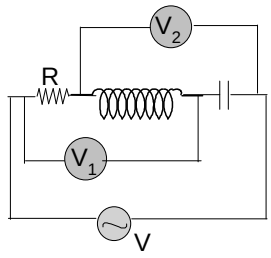
7. A wire is bent to form semi-circle of radius a. The wire rotates about its one end with angular velocity ω. Axis of rotation is perpendicular to plane of the semicircle. In the space, a uniform magnetic field of induction B exists along the axis of rotation as shown in figure. Then



(A) Potential difference between P and Q is equal to $2B\omega a^2$
(B) Potential difference between P and Q is equal to $2\pi^2 B\omega a^2$
(C) P is at higher potential Q
(D) P is at lower potential than Q

Space for Rough work

8. In an R-L-C series circuit shown in the figure, the readings of voltmeters V_1 and V_2 are 100V and 120V, respectively. The source voltage is 130V. For this situation, mark out the correct statement(s).



(A) Voltage across resistor, inductor and capacitor are 50V, $50\sqrt{3}$ V and $120+50\sqrt{3}\,\text{V}$, respectively

(B) Voltage across resistor, inductor and capacitor are 50V, $50\sqrt{3}$ V and $120-50\sqrt{3}\,\text{V}$, respectively

(C) Power factor of the circuit is $\frac{5}{13}$

(D) The circuit is capacitive in nature

9. An α-particle of mass m_α and q_α moves a velocity $V_0=\left(2\hat{i}+3\hat{j}-\hat{k}\right)\text{ms}^{-1}$ through a region, where both a uniform magnetic field and a uniform electric field. Which of the following statements is/are correct?

[Given, $B=\left(2i+4j+k\right)\text{T}$ and $E=\left(4i-j-2k\right)\text{VM}^{-1}$]

(A) Acceleration of α-particle is perpendicular to Z-axis

(B) Acceleration of α-particle is perpendicular to Z-axis

(C) de-Broglie's wavelength associated with α-particle is increasing with time

(D) de-Broglie's wavelength of α-particle keeps on decreasing with time

Space for Rough work

10. The polonium isotope $^{210}_{84}\text{Po}$ is unstable, it emits an α-particle with energy of 5.30MeV. The atomic mass of $^{210}_{84}\text{Po}$ is 209.9829 u and atomic mass of $^{4}_{2}\text{He}$ is 4.0026 u. Then,

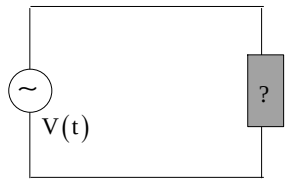
(A) Daughter nucleus is $^{206}_{82}\text{Pb}$

(B) disintegration energy of α-decay is 5.30MeV.

(C) disintegration energy of α-decay is 5.40MeV.

(D) mass of daughter nucleus is 205.9745u.

11. An AC voltage source is connected to a "black box" which contains a circuit, as shown in Figure. The elements in the circuit and their arrangement, however, are unknown. Measurements outside the black box provide the following information



V(t) = (80V) sin ωt

I(t) = (1.6A) sin (ωt+π/4)

Based on the above information we can infer:

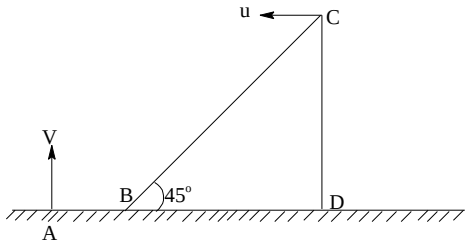
(A) The circuit is largely capacitative

(B) The circuit in the black box is in resonance

(C) The box must contain at least one resistor and one capacitor

(D) The average power delivered to the black box by the AC source is 45.25 W

12. A large wedge BCD, having its inclined surface at an angle $\theta = 45^o$ to the horizontal, is travelling horizontally leftwards with uniform velocity $u=10\text{m/s}$. At some instant a particle is projected vertically up with speed $V=20\text{m/s}$ from point A on ground lying at some distance right to the lower edge b of the wedge. The particle strikes the incline BC normally, while it was falling. $[g=10\text{m/s}^2]$



(A) The distance AB at the instant the particle was projected from A is 15m

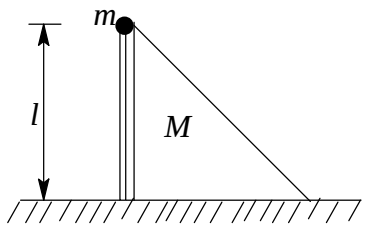
(B) The distance of lower edge B of the wedge from point A at the instant particle strikes the incline is 15m

(C) The distance of lower edge B of the wedge from point A at the instant particle strikes the incline is 30m

(D) The path of the particle in the reference frame attached to the wedge is parabolic

Space for Rough work

13. A weightless rod of length l with a small load of mass m at one of its end is held vertical with its lower end hinged on a horizontal surface. The load touches a wedge right sets the system in motion (see figure), with rod rotating freely in vertical plane about its lower end. There is no friction. If the rod forms an angle $\theta = \pi/3$ with the vertical at the moment the load separates from the wedge, then



(A) The mass ratio $\frac{M}{m}$ =3

(B) The speed of the wedge at that moment will be $\sqrt{\frac{gl}{8}}$

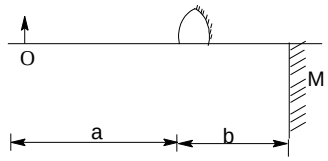
(C) The mass ratio $\frac{M}{m}$ =4

(D) The speed of the wedge at that moment will be $\sqrt{\frac{gl}{2}}$

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

14. In figure, L is half part of an equiconvex glass lens $(\mu = 1.5)$ whose surface have radius of curvature $R = 40cm$ and its right surface is silvered. Normal to its principle axis a plane mirror M is placed on right of the lens. Distance between lens L and mirror M is b. A small object O is placed on left of the lens such that there is no parallax between final images formed by the lens and mirror. If transverse length of final image formed by lens is twice that of image formed by the mirror, calculate distance 'a' in cm between lens and object.

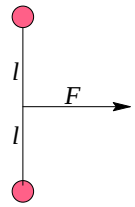


Space for Rough work

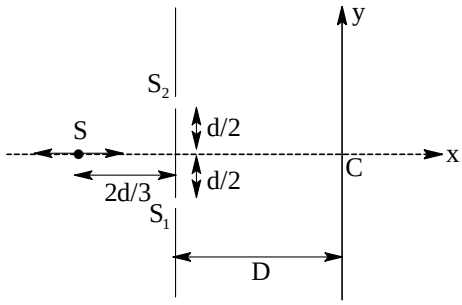
15. Binding energy per nucleon of a fixed nucleus X^A is 6Me V. It absorbs a neutron moving with $KE=2MeV$,and converts into Y, emitting a photon of energy $1\ MeV$ .If the Binding Energy per nucleon of Y (in MeV)is found to be $\frac{(xA-1)}{(A+1)}$. Find the value of x.

16. A neutral atom of atomic mass number 100 which is stationary at the origin in gravity free emits an α -particle (A) in z-direction. The product ion is P. A uniform magnetic field exists in the x-direction. Disregard the electromagnetic interaction between A and P. If the angle of rotation of A after which A and P will meet for the first time is $6n\pi/25$ radius, what is the value of n?

17. Two identical masses, placed on a smooth horizontal surface, are connected by a light string of length 2l. A constant force F starts acting on the system in the direction shown. After some time the masses collide inelastically. The energy lost in this collision is Fl/N . Find the value of N.



18. In Young's Double slit experiment, a symmetrically placed, long, linear and monochromatic source (S) of constant power is undergoing oscillations along x-axis with small amplitude A. Its mean position is shown in the figure. If I_0 is the intensity due to single slit near centre (C) of screen (y-z plane) when source is its mean position. If the amplitude of intensity variation ΔI_{max} at any of the bright fringe near centre is $\frac{8xAI_0}{25d}$, then find the value of x. [Assume D>>d]



Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **5 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. Two solid compounds 'X' and 'Y' dissociates at certain temperature as follows:

$X(s) \rightleftharpoons A(g) + 2b(g)$ $Kp_1 = 9 \times 10^{-3}$ atm^3

$Y(s) \rightleftharpoons 2B(g) + C(g)$ $Kp_2 = 4.5 \times 10^{-3}$ atm^3.

The total pressure (in atm) of gases over a mixture of 'X' and 'Y' is

(A) 1.5 (B) 0.45
(C) 0.6 (D) 0.9

20. The potential of an indicator electrode versus the standard hydrogen electrode, which originally contains 0.1 M MnO_4 and 0.8 M H^+ and which was treated with Fe^{2+} necessary to reduce 90% of MnO_4^- to Mn^{2+} ($E^o_{MnO_4^-/Mn^{2+}} = 1.51\,V$), is

(A) 1.41 V (B) 2.42 V
(C) 1.04 V (D) 0.42 V

21. For the reaction $\frac{1}{2}N_2(g) + \frac{3}{2}H_2(g) \rightarrow NH_3(g)$, $\Delta G^0_{298K} = -16.45$ KJ/mol and $\Delta H^0_{298K} = -46.11$ KJ/mol. Then the value of $\Delta G^0_{500\,K}$ is

(Assume $\Delta H°$ is independent of temperature).

(A) 3.66 KJ/mol (B) –12.79 KJ/mol
(C) –20.11 KJ/mol. (D) –10.55 kJ/mol

Space for Rough work

22. Match the following:

	Column I		Column II
P.	All except one have ns^2 configuration	1.	3d series transition metals
Q.	Only 3 elements have ns^2 configuration and	2.	4d series transition metals
R.	Only 2 elements with ns^1 configuration	3.	5d series transition metals
S.	Only 1 element with ns^0 configuration	4.	6d series transition metals

(A) P-4;Q-2,3;R-1;S-3,1 (B) P-4;Q-2,3;R-1,3;S-2

(C) P-4;Q-2;R-1,3;S-2 (D) P-2;Q-3;R-4;S-1,2

23. The distance between any two I atoms in BI_3 is found to be 3.54 Å. What is the covalent radius of boron, if covalent radius of I atom is 1.33 Å? (Assume all bonds are single bonds in BI_3)

(A) 0.71 Å (B) 1.77 Å

(C) 1.2 Å (D) 1.34 Å

(One or More than one correct type)

This section contains **EIGHT** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

24. Which is/are correct

(A) Except Sc, among 3d series transition metals all can show $+2$ oxidation state.

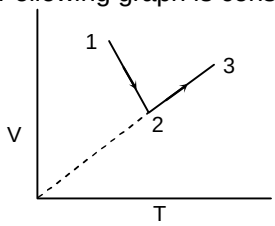
(B) Sc (I) is more stable than Sc(II) and Sc(III)

(C) Ti (IV) is more stable than Ti(III) and Ti(II)

(D) Ferrates $[FeO_4]^{2-}$ are not stable, so decompose to give Fe_2O_3 and O_2.

Space for Rough work

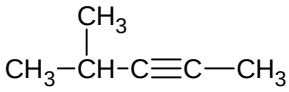
25. Following graph is constructed for the fixed amount of gas

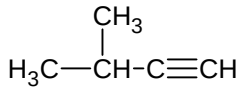


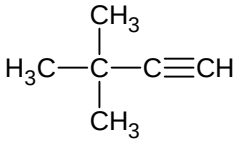
(A) From 1–2 pressure will decrease
(B) From 2–3 pressure remains constant
(C) Gas pressure at (3) is greater than at state (1)
(D) From 1–2 pressure will increase.

26. Which is /are correct
(A) Cr(VI) is more stable than Mo((VI) and W(VI)
(B) Sc does not exhibit variable oxidation states.
(C) The transition metal ions containing empty d-orbital's $[e.g\,Sc^{3+}\text{ and }Ti^{+4}\text{ions}]$ or completely filled d-orbital's $[e.g\,Cu^{+},Ag^{+},Zn^{2+},Cd^{2+},Hg^{2+}\text{etc}]$ in dry state are colourless
(D) $E^{o}_{Cu^{2+}/Cu}$ is +ve among $E^{o}_{M^{2+}/M}$ of 3d series elements as Energy required to transform $Cu(s) \rightarrow Cu^{2+}(g)$ is not balanced by hydration enthalpy.

27. Which of the following compounds gives optically inactive compound on isomerization when treated with sodamide in liquid NH_3 or with Na/NH_3.

(A) 

(B) 

(C) 

(D) $H_3C–C\equiv C–CH_3$

Space for Rough work

28. Buffer solution A of a weak mono protic acid and its sodium salt in the concentration ratio x : y has pH = (pH)1. Buffer solution B of same acid and its sodium Salt in the concentration ratio y : x has pH = $(pH)_2$ If $(pH)_2 - (pH)_1 = 1$ unit and $(pH)_1 + (pH)_2 = 9.5$ units then

(A) pKa = 4.75 (B) $\frac{x}{y} = 2.36$

(C) $\frac{x}{y} = 3.162$ (D) pKa = 5.25

29. In a hypothetical reaction $A \rightarrow B$, the activation energies for the forward and backward reactions are 15 and 9 kJ/mol respectively. The potential energy of A is 10 kJ/mol, then which is/are correct
(A)the threshold energy of the reaction is 25 kJ/mol
(B)potential energy of B is 16 kJ
(C)heat of reaction is 6 kJ
(D)the reaction is endothermic

30. van't Hoff equation is

(A) $(d/dT)\ln K = \left(-\Delta E/RT^2\right)$ (B) $(d/dT)\ln K = +\left(E/RT^2\right)$

(C) $(d/dT)\ln K = -(\Delta E/RT)$ (D) $K=Ae^{-E/RT}$

31. For ideal gases which options are correct?

(A) Entropy change for isobaric (at constant pressure): $\Delta S = 2.303 nC_p \log\left(\frac{T_2}{T_1}\right)$

(B) Entropy change for isobaric (at constant pressure): $\Delta S = 2.303 nC_p \log\left(\frac{V_2}{V_1}\right)$

(C) Entropy change for isochoric (at constant volume): $\Delta S = 2.303 nC_v \log\left(\frac{T_2}{T_1}\right)$

(D) Entropy change for isochoric (at constant volume): $\Delta S = 2.303 nC_v \log\left(\frac{P_2}{P_1}\right)$

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

32. A cation exchange regin RH_2 can replace Ca^{+2} in hard water as $RH_2 + Ca^{+2} \rightarrow RCa + 2H^+$. 1 L of Hard water (Hardness x ppm) after passing through RH_2 has pH = 2. Then the value of $\frac{x}{40}$ is.

33. $Ore + Na_2CO_3 \xrightarrow[\Delta]{air} \text{obtained solid} + \text{Gas A}$

after cooling and filtered

Fillrate

E $\xleftarrow{KCl}$ D $\xleftarrow{\text{acidification}}$ B

C
residue

Gas A $\rightarrow$ Line water test +ve

$\xrightarrow{\text{with B}}$ can't be oxidised

E is an oxidising agent and when a paper dipped in E solution comes in contact with SO_2 changes colour from orange to green.

Now, how many are correct about above flow diagram?

(A) Lime is added with Na_2CO_3, keeps the mass porous so that air becomes accessible to all parts reaction mixture and prevents fusion

(B) 'C' is mainly Fe_2O_3

(C) Filtrate is a yellow solution.

(D) During acidification a redox reaction is taking place.

(E) D contain 2 salts in solution and when fractional crystallisation occurs salt of lower molecular mass comes out first and is filtered off.

(F) Acidication gives two salts both hydrated. The one which has more water of hydration is less soluble

(G) When pH of solution is increased the colour of solution of E changes from orange to yellow

(H) The oxidising anions present in flow diagram can be distinguished by $AgNO_3$ One gives reddish-brown ppt and other gives brick red ppt.

Space for Rough work

34. How many of the following contain at least one tetrahedral unit formed by atoms.
$CrO_4^{2-}, K_2Cr_2O_7, KMnO_4, K_2MnO_4, Mn_2O_7, K_3VO_4$, pyrohosphoric acid, pyrosulpharic acid, Chloric acid, perchloric acid.

35. How many of the following are amphoteric
$V_2O_5, V_2O_3, CrO, Cr_2O_3, Mn_2O_7, ZnO$

36. An Ion 'M^{a+}', (at no : 26) has paramagnetic moment of 4.8 BM. Then the value of 'n' is

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **5 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. Image of the curve $xy=1$ in the curve $\left(x+\sqrt{x^2+1}\right)\left(y+\sqrt{y^2+1}\right)=1$ (on coordinate plane) is

(A) $xy=1$ (B) $xy+1=0$

(C) $xy=0$ (D) $x^2+y^2=1$

38. In a triangle ABC (with usual notations), if ex-radii r_1, r_2, r_3 are in H.P., then $\frac{a+c}{b}$ is-

(A) 1 (B) 2

(C) 3 (D) 4

39. If A is an invertible square matrix of order n such that $|\text{adj}(\text{adj } A)| = |A|^{(2n^2-7n+7)}$, then sum of possible values of n is-

(A) 3 (B) 4

(C) 5 (D) 6

40. Let $f(x)$ is an even function such that $f(x)+f(x-3)=x(x-3)+1$, then $\int_0^3 \frac{f(x)dx}{(x^2-3x+1)}$ is

(A) $-\frac{2}{3}$ (B) $-\frac{3}{2}$

(C) $\frac{2}{3}$ (D) $\frac{3}{2}$

Space for Rough work

41. The range of function $f(x)=e^{\left(\sin^{-1}x+\frac{\cos^{-1}x}{4}\right)}$ is

(A) $\left[e^{\frac{\pi}{4}},e^{\frac{\pi}{2}}\right]$ (B) $[0,1]$

(C) $\left[e^{-\frac{\pi}{2}},e^{\frac{\pi}{2}}\right]$ (D) $\left[e^{-\pi},e^{\pi}\right]$

(One or More than one correct type)

This section contains **EIGHT** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

42. Let S_n denotes the sum of first n term of an arithmetic progression whose first term is -4 and common difference is 1. If $V_n=2S_{n+2}-2S_{n+1}+S_n(n\in N)$, then

(A) $V_n=\frac{-9n^2+5n-12}{2},(n\in N)$ (B) $V_n=\frac{n^2+5n+12}{2},(n\in N)$

(C) Minimum value of V_n is -9 (D) Minimum value of V_n is $-\frac{73}{8}$

43. The value of x satisfying the equation $x^9+\frac{9}{8}x^6+\frac{27}{64}x^3-x+\frac{219}{512}=0$ is-

(A) $\frac{-1+\sqrt{13}}{4}$ (B) $-\frac{1}{2}$

(C) $\frac{-1-\sqrt{13}}{4}$ (D) more than two real and distinct values

Space for Rough work

44. If α, β, γ are roots of $x^3 + 2x^2 - 3x + 1 = 0$, then value of $\frac{\alpha\beta}{\alpha+\beta} + \frac{\alpha\gamma}{\alpha+\gamma} + \frac{\beta\gamma}{\beta+\gamma}$ is less than

(A) 2 (B) 3
(C) 4 (D) 5

45. The function $f(x)$ is defined by $f(x) = \frac{ax+b}{cx+d}$, where $a, b, c \text{ and } d$ are non-zero real number such that $f(19) = 19, f(97) = 97 \text{ and } f(f(x)) = x$ for all values except $-\frac{d}{c}$. If range of $f(x)$ is $R - \{\alpha\}$ (where R denotes real number set), than

(A) number of proper divisors of α is 2 (B) number of proper divisors of α is 3
(C) sum of all divisors of α is 180 (D) sum of all divisors of α is 90

46. It is given that there are two sets of real numbers $A \equiv \{a_1, a_2, a_3, ...a_{100}\} \text{and } \{b_1, b_2, b_3, ...b_{50}\}$. If $f : A \to B$ is such that every element in B has an inverse image and $f(a_1) \le f(a_2) \le ..... \le f(a_{100})$, then the number of such mapping is-

(A) ${}^{100}C_{51}$ (B) ${}^{99}C_{50}$
(C) ${}^{100}C_{49}$ (D) ${}^{99}C_{49}$

47. The real numbers x_1, x_2, x_3 satisfying the equation $x^3 - x^2 + bx + c = 0$ are in A.P. Which of the following are necessary?

(A) $b \le \frac{1}{3}$ (B) $c \ge -\frac{1}{27}$
(C) $b \le -\frac{1}{3}$ (D) $c \ge \frac{1}{27}$

Space for Rough work

48. The solution of differential equation $3\frac{dx}{dy}=\frac{x}{x^3-y}$ is $x^{\ell}=mx^n y+c$, then which of the following is/are **CORRECT**?{c is any arbitrary constant}

(A) $\ell+m+n=11$ (B) $\ell+n=9$

(C) $\ell+2m=10$ (D) $m+n=4$

49. All the 7-digit Numbers containing each of the digits 1, 2, 3, 4, 5, 6, 7 exactly once, and not divisible by 5, are arranged in increasing order. Then which of the following is/are **CORRECT**?

(A) 1993rd number in the list is 4312567 (B) 1996th number in the list is 4312756

(C) 2000th number in the list is 4315672 (D) 1999th number in the list is 43152763

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. Let z_1, z_2 and z_3 be complex numbers such that $|z_1|=|z_2|=|z_3|=|z_1+z_2+z_3|=2$. If $|z_1-z_2|=|z_1-z_3|$ *and* $z_2 \neq z_3$. Then find the value of $|z_1+z_2||z_1+z_3|$.

51. If $\lim_{x\to 1}\frac{100}{1-x^{100}}-\frac{50}{1-x^{50}}=5A$; find the value of A.

52. If $\sum_{i=0}^{\infty}\sum_{j=0}^{\infty}\sum_{k=0}^{\infty}\frac{1}{3^i 3^j 3^k}=\frac{27\times A}{208}$; where A is single digit natural, then find the value of A.

53. Let X be the set of all positive integer greater than or equal to 8 and let $f:X\to X$ be a function such that $f(x+y)=f(xy)$ for all $x\geq 4, y\geq 4$ .if $f(8)=9$ .Determine $f(9)$

54. The three different polynomials x^2+ax+b, x^2+x+ab and ax^2+x+b have exactly one common zero. Where a, b are non-zero real numbers. Find the value of $a+2b$.

Space for Rough work